Week 9 Assignment

Correlation and linear regression

Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas before Week 10's lecture (Monday). See the syllabus for the late policy.

Using the built-in `trees` dataset:

1. Compute the correlation between `Height` and `Volume`, with a significance test.
2. Fit a simple linear regression of `Volume ~ Height`.
3. Report the slope, R^2, and p-value, and interpret the slope in plain language (what does it mean biologically?).
4. Check the regression diagnostics.

Does a straight line look like a reasonable fit here?

5. In 2–3 sentences: `trees` is observational data. What would need to be true about how the data were collected for you to make a causal claim ("increasing height causes increased volume") instead of just a correlational one?